

Al-Aqsa University

Faculty of Computers and IT

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جامعة الأقصى
كلية الحاسبات وتكنولوجيا المعلومات
قسم الهندسة والشبكات

Operating Systems for Intelligent Systems (ENGS2306)

Lecture 1: Introduction to Linux

Lecturer: Eng. Mohammed E. Alhams

Semester: Second Semester 2024/2025

Welcome to the Course

- **Course:** Operating Systems for Intelligent Systems (ENGS2306)
- **Lecturer:** Eng. Mohammed E. Alhams
- **Duration:** 14 weeks
- **Delivery:** Pre-recorded lectures via YouTube and Moodle
- **Week 1 Objectives:**
 - Install Ubuntu 20.04 LTS in VMware Workstation Player.
 - Explore Linux command-line commands for system navigation and management.
 - Understand Linux's critical role in intelligent systems.
- **Why This Lecture?:**
 - Establishes the foundation for hands-on Linux exploration.
 - Introduces virtualization with VMware for safe OS experimentation.
 - Prepares students for practical tasks throughout the course.

What is Linux?

- **Definition:** Linux is an open-source operating system kernel, forming the core of distributions like Ubuntu, Fedora, and Raspberry Pi OS.
- **History:**
 - Created by Linus Torvalds in 1991.
 - Built on Unix principles but freely available and community-driven.
 - Evolved into a global standard for servers, mobile devices, and embedded systems.
- **Key Features:**
 - **Open-Source:** Code is accessible, modifiable, and free.
 - **Multi-user & Multitasking:** Supports multiple users and simultaneous processes.
 - **Modular:** Kernel manages hardware; user-space tools handle tasks like file management.
 - **Secure:** Robust permission systems and regular updates.
- **Relevance to Intelligent Systems:**
 - Powers Android devices, IoT gadgets, and smart appliances.
 - Lightweight and customizable for resource-constrained environments.
 - Provides tools for managing processes, memory, and I/O in intelligent systems.

Linux in Intelligent Systems

- **Applications:**
 - **Mobile Devices:** Android uses the Linux kernel for process scheduling and hardware management.
 - **IoT Devices:** Linux runs on smart thermostats, cameras, and sensors (e.g., Raspberry Pi OS).
 - **Embedded Systems:** Used in automotive systems, medical devices, and industrial controllers.
- **Why Linux?:**
 - **Scalable:** Runs on low-power devices to high-performance servers.
 - **Customizable:** Kernel and tools can be tailored for specific tasks.
 - **Community Support:** Extensive documentation and forums (e.g., Stack Overflow, Ubuntu forums).
- **Course Relevance:**
 - Learn Linux internals to understand OS management in intelligent systems.
 - Practice commands to interact with system resources directly.

Linux Architecture Overview

- **Components:**
 - **Kernel:** Manages CPU, memory, I/O, and device drivers.
 - **Shell:** Command-line interface (e.g., Bash) for user interaction with the kernel.
 - **File System:** Hierarchical structure (/home, /etc, /var) for data organization.
 - **Utilities:** Tools like ls, grep, top for system management.
- **How It Works:**
 - Kernel communicates with hardware (e.g., CPU, disk).
 - Shell interprets user commands (e.g., ls) and passes them to the kernel.
 - Utilities provide user-friendly ways to manage files, processes, and more.
- **Example:**
 - Running `ls /home` lists files in the user's home directory, leveraging the file system and kernel.
- **Why It Matters:** Understanding architecture enables efficient resource management in intelligent systems.

Why Use VMware?

- **Purpose:** VMware Workstation Player is a free virtualization tool to run multiple OSs (e.g., Ubuntu, Raspberry Pi OS) on one computer.
- **Benefits:**
 - Safe experimentation: Test Linux without risking your main system.
 - Simulates real-world OS deployment (e.g., servers, embedded devices).
 - Supports course tasks: Ubuntu VMs (Weeks 1–11), Raspberry Pi OS (Week 12).
- **Key Features:**
 - Create and manage virtual machines (VMs) with customizable CPU, RAM, and storage.
 - Snapshot feature to save and revert VM states.
 - Portable: Run VMs on any compatible computer.
- **Course Role:** Provides a controlled environment to explore Linux internals.

Preparing for Ubuntu Installation

- **System Requirements:**

- Computer with 4GB RAM (8GB recommended), 20GB free storage.
- Stable internet for downloading Ubuntu ISO (or Moodle link).

- **Resources Needed:**

- Ubuntu 20.04 LTS ISO: Download from Ubuntu official website or Moodle.
Follow this link to download the desktop ISO image: <https://shorturl.at/mQ1ZG>
- VMware Workstation Player: Available via Moodle or VMware website.

- **Pre-Installation Checklist:**

- Verify system specs (RAM, storage, CPU).
- Ensure internet access for downloading (use stable hours in Gaza).
- Have a USB drive as a backup for ISOs if internet is unreliable.

- **Note:** Contact Eng. Mohammed E. Alhams via Moodle for setup issues.

Installing Ubuntu 20.04 LTS in Vmware

- **Step-by-Step Guide:**

1. Download Ubuntu 20.04 LTS ISO from Ubuntu website or Moodle.
2. Open VMware Workstation Player and select "Create a New Virtual Machine."
3. Choose "Installer disc image file (iso)" and select the Ubuntu ISO.
4. Select "Linux" and "Ubuntu 64-bit" as the OS type.
5. Configure VM: Allocate 2 CPUs, 4GB RAM, 20GB disk space.
6. Start the VM and follow the Ubuntu installer.
7. Select "Erase disk and install Ubuntu" (VM disk, not physical).
8. Complete installation and log into Ubuntu.

- **Tips:**

1. Save the ISO to a local drive for reuse.
2. Take a VMware snapshot after installation to revert if needed.